Chapter 2

p.2

Example 1.

$\{a_t\}$ is a white noise with $E(a_t)=0$ and $V(a_t)=\sigma^2$.

$Z_t = a_t + 0.5\,a_{t-1}$ is a time series

with $Z_{t-k} = a_{t-k} + 0.5\,a_{t-k-1}$

$$E(Z_t) = E(a_t + 0.5\,a_{t-1}) = E(a_t) + 0.5\,E(a_{t-1}) = 0$$

$$V(Z_t) = V(a_t + 0.5\,a_{t-1}) = V(a_t) + 0.5^2\,V(a_{t-1})$$

$$= \sigma^2 + 0.25\,\sigma^2 = 1.25\,\sigma^2$$

$$Cov(Z_t, Z_{t-k}) = E(Z_t\,Z_{t-k}) - E(Z_t)E(Z_{t-k})$$

$$= E(Z_t\,Z_{t-k})$$

$$= E\left[(a_t + 0.5\,a_{t-1})(a_{t-k} + 0.5\,a_{t-k-1})\right]$$

$$= E\left[a_t a_{t-k} + 0.5\,a_{t-1}a_{t-k} + 0.5\,a_t a_{t-k-1} + 0.5^2 a_{t-1}a_{t-k-1}\right]$$

$$= \begin{cases} 1.25\,\sigma^2 & \text{if } k=0, \\ 0.5\,\sigma^2 & \text{if } k=1. \end{cases}$$

N.B. If $k=0$, $E(Z_t^2) = V(Z_t) = 1.25\,\sigma^2$

OR $= E\left[a_t a_t + 0.5\,a_{t-1}a_t + 0.5\,a_t a_{t-1} + 0.25\,a_{t-1}^2\right]$

$= \sigma^2 + 0.25\,\sigma^2 = 1.25\,\sigma^2.$

If $k=1$, $E(Z_t Z_{t-1}) = E\left[a_t a_{t-1} + 0.5\,a_{t-1}^2 + 0.5\,a_t a_{t-2} + 0.25\,a_{t-1}a_{t-2}\right]$

$= 0.5\,\sigma^2$

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{0.5\,\sigma^2}{1.25\,\sigma^2} = 0.4.$$

P.4 Example 2

$Y_t = 5 + 2t + X_t$, where X_t is zero-mean stationary series with autocovariance function γ_k.

(a) $E(Y_t) = E(5 + 2t + X_t) = 5 + 2t + E(X_t)$

$= 5 + 2t$ since $E(X_t) = 0$.

(b) $E\{[Y_t - E(Y_t)][Y_{t-k} - E(Y_{t-k})]\}$

$= E\{[5 + 2t + X_t - 5 - 2t][5 + 2(t-k) - 5 - 2(t-k)]\}$

$= E(X_t X_{t-k}) = \gamma_k$

(c) Is Y_t stationary?

No, since $E(Y_t) = 5 + 2t$, which depends on t.

∴ Y_t is NOT stationary.

Example 3

$Z_t = A \sin(\omega t + \theta)$

where A is a random variable with $E(A) = 0$ and $V(A) = 1$.

θ is another independent random variable to A, $\theta \sim U[-\pi, \pi]$.

$E(Z_t) = E[A \sin(\omega t + \theta)]$

$= E(A) E[\sin(\omega t + \theta)]$

$= 0 \cdot E[\sin(\omega t + \theta)]$

$= 0.$

N.B. $E[\sin(\omega t + \theta)]$

$= E[\sin\omega t \cos\theta + \cos\omega t \sin\theta]$

$= \sin\omega t\, E(\cos\theta) + \cos\omega t\, E(\sin\theta)$

$= \sin\omega t \int_{-\pi}^{\pi} \cos\theta\, d\theta + \cos\omega t \int_{-\pi}^{\pi} \sin\theta\, d\theta$

$= \sin\omega t \sin\theta \Big|_{-\pi}^{\pi} + \cos\omega t \,(-\cos\theta)\Big|_{-\pi}^{\pi})$

$= \sin\omega t\,(0) + \cos\omega t\,(0)$

$= 0.$

$$E(Z_t Z_{t+k}) = E\left[A\sin(\omega t+\theta)\cdot A\sin[\omega(t+k)+\theta] \right] \quad \text{p.5 top}$$
$$= E\left[A^2 \sin(\omega t+\theta)\sin[\omega(t+k)+\theta] \right]$$

Consider $\cos(B-A) = \cos B\cos A + \sin B\sin A$

$\Rightarrow \cos(A+B) = \cos A\cos B - \sin A\sin B$

$$\cos(B-A) - \cos(A+B) = 2\sin A\sin B$$

$$\therefore \sin A\sin B = \frac{1}{2}\left[\cos(B-A) - \cos(A+B)\right]$$

where $\left.\begin{aligned} A &= \omega t+\theta \\ B &= \omega(t+k)+\theta \end{aligned}\right\} \Rightarrow \begin{aligned} B-A &= \omega k \\ A+B &= \omega(2t+k)+2\theta \end{aligned}$

$$\therefore \sin(\omega t+\theta)\sin[\omega(t+k)+\theta]$$
$$= \frac{1}{2}\left\{ \cos(\omega k) - \cos[\omega(2t+k)+2\theta] \right\}$$

$$E(Z_t Z_{t+k}) = E(A^2)\, E\left\{ \frac{1}{2}\left[\cos(\omega k) - \cos[\omega(2t+k)+2\theta]\right] \right\}$$

$$= 1\cdot\frac{1}{2}\cos(\omega k) - \frac{1}{2}E\left\{ \cos[\omega(2t+k)]\cos 2\theta - \sin[\omega(2t+k)]\sin 2\theta \right\}$$

$$= \frac{\cos(\omega k)}{2} - \frac{1}{2}\left[\cos[\omega(2t+k)]\, E(\cos 2\theta) - \sin[\omega(2t+k)]\, E(\sin 2\theta) \right]$$

$$= \frac{\cos(\omega k)}{2} - \frac{1}{2}\left[\cos[\omega(2t+k)] \int_{-\pi}^{\pi} \cos 2\theta \frac{1}{2\pi} d\theta - \sin[\omega(2t+k)] \int_{-\pi}^{\pi} \sin 2\theta \frac{1}{2\pi} d\theta \right]$$

$$= \frac{\cos(\omega k)}{2} - \frac{1}{2}\cdot\frac{1}{2\pi}\left[\cos[\omega(2t+k)] \left.\frac{\sin 2\theta}{2}\right|_{-\pi}^{\pi} - \sin[\omega(2t+k)] \left.\frac{-\cos 2\theta}{2}\right|_{-\pi}^{\pi} \right]$$

$$= \frac{\cos(\omega k)}{2} - \frac{1}{4\pi}\left[\cos[\omega(2t+k)]\times 0 - \sin[\omega(2t+k)]\times 0 \right]$$

$$= \frac{\cos(\omega k)}{2} - 0 = \frac{1}{2}\cos(\omega k)$$

Since the mean and covariance of Z_t do not depend on t, hence Z_t is stationary.

2.2.2 Sample Autocorrelation (p.5 middle)

For a series $\{X_t : t = 1, 2, \ldots, n\}$, define

$$C_k = \sum_{t=k+1}^{n} X_t X_{t-k} - (n-k)\bar{X}^2, \quad k = 0, 1, 2, \ldots.$$

Then

$$r_k = \frac{C_k}{C_0}$$ is the kth sample autocorrelation coefficient of $\{X_t\}$. A plot of r_k Vs k is called the correlogram of $\{X_t\}$. C_k is the sample autocovariance and C_0 is sample variance of $\{X_t\}$.

N.B.

(i) Limits "$\pm 2 \times \frac{1}{\sqrt{n}}$" [$\pm 2\sigma$, similar to confidence interval] are conventionally used to assess significantly non-zero r_k.

(ii) Slowing decaying r_k, especially with $r_1 \approx 1$, is indicative of non-stationarity in $\{X_t\}$. May need to eliminate trend.

(iii) Sudden peaks in r_k suggest seasonal effects, again eliminate whenever possible.

(iv) Removal of trend or seasonal effect can induce auto-correlation pattern in the residual series.

(v) The autocovariance function of $\{X_t\}$ is:

$$Cov(X_t, X_{t-k}) = E\{[X_t - E(X_t)][X_{t-k} - E(X_{t-k})]\}.$$

The autocorrelation function (ACF) of $\{X_t\}$ is:

$$Corr(X_t, X_{t-k}) = Cov(X_t, X_{t-k}) \Big/ [V(X_t)\, V(X_{t-k})]^{1/2}.$$

P.6 Example 4

Reconsider Example 1 (on p.①),

$$Z_t = a_t + 0.5\, a_{t-1}$$

this is an MA1 process, with lag-1. The theoretical autocorrelation coefficient (of lag-1) is

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{0.5\sigma^2}{1.25\sigma^2} = 0.4.$$

2.3 General Linear Process

A general linear process (GLP) $\{Y_t\}$ is one that can be represented by a linear combination of present and past white noise terms, namely

$$Y_t = \psi_0 e_t + \psi_1 e_{t-1} + \psi_2 e_{t-2} + \cdots = \sum_{i=0}^{\infty} \psi_i e_{t-i}$$

Without loss of generality, ψ_0 is usually set to be 1. For the right hand side of the series to be convergent, we need the conditions on the ψ_i terms. For our purpose, we need to impose the condition

$$\sum_{i=1}^{\infty} \psi_i^2 < \infty$$

that is: $\sum_{i=1}^{\infty} \psi_i^2$ is finite.

Theorem: A general linear process with condition $\sum_{i=1}^{\infty} \psi_i^2 < \infty$ is stationary.

Proof: $E(Y_t) = E\left(\sum_{i=0}^{\infty} \psi_i e_{t-i}\right) = \sum_{i=0}^{\infty} \psi_i E(e_{t-i})$

$$= \sum_{i=0}^{\infty} \psi_i (0) = 0.$$

$$V(Y_t) = V\left(\sum_{i=0}^{\infty} \psi_i e_{t-i}\right) = \sum_{i=0}^{\infty} \psi_i^2 V(e_{t-i})$$

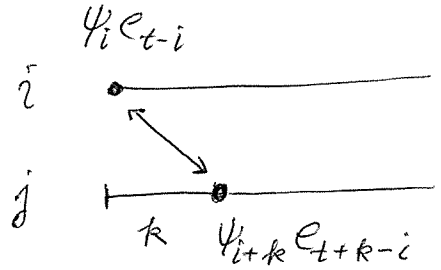
$$= \sum_{i=0}^{\infty} \psi_i^2 (\sigma_e^2) = \sigma_e^2 \sum_{i=0}^{\infty} \psi_i^2 < \infty$$

which is free of t.

$$Cov(Y_t, Y_{t+k}) = E(Y_t Y_{t+k}) - E(Y_t)E(Y_{t+k})$$

$$= E(Y_t Y_{t+k}) = E\left(\sum_{i=0}^{\infty} \psi_i e_{t-i} \sum_{j=0}^{\infty} \psi_j e_{t+k-j}\right)$$

$$= E\left(\sum_{i=0}^{\infty} \psi_i \psi_{i+k} e_{t-i} e_{t+k-i}\right)$$



$$= \sum_{i=0}^{\infty} \psi_i \psi_{i+k} E(e_{t-i} e_{t+k-i})$$

$$= \sum_{i=0}^{\infty} \psi_i \psi_{i+k} \sigma_e^2$$

$$= \sigma_e^2 \sum_{i=0}^{\infty} \psi_i \psi_{i+k} < \infty$$

Since the mean, variance and autocovariance all do not depend on t, i.e., Y_t is stationary.

2.4 Exponential Smoothing for a Linear Trend Process

Suppose our process is of the form

$$x_t = b_1 + b_2 t + \varepsilon_t, \quad \text{where } \varepsilon_t \sim NID(0, \sigma^2),$$

if we use simple exponential smoothing, then recall [P.5, section 1.4.1]

$$S_T = \alpha \sum_{k=0}^{T-1} (1-\alpha)^k x_{T-k} + (1-\alpha)^T S_0, \quad \text{let } \beta = 1-\alpha$$

$$= \alpha \sum_{k=0}^{T-1} \beta^k x_{T-k} + \beta^T S_0$$

$$\Rightarrow E(S_T) = E\left[\alpha \sum_{k=0}^{T-1} \beta^k x_{T-k} + \beta^T S_0\right]$$

$$= \alpha \sum_{k=0}^{T-1} \beta^k E(x_{T-k}) + \beta^T S_0$$

$$= \alpha \sum_{k=0}^{T-1} \beta^k \left[b_1 + b_2 (T-k)\right] + \beta^T S_0$$

As T approaches to ∞, we have $\lim_{T\to\infty} \beta^T S_0 = 0$ and

$$E(S_T) = (b_1 + b_2 T)\alpha \sum_{k=0}^{\infty} \beta^k - b_2 \alpha \sum_{k=0}^{\infty} k\beta^k$$

Notice that

① $\sum_{k=0}^{\infty} \beta^k = 1 + \beta + \beta^2 + \cdots \infty = \frac{1}{1-\beta} = \frac{1}{1-(1-\alpha)} = \frac{1}{\alpha}$

② $\sum_{k=0}^{\infty} k\beta^k = \beta \sum_{k=0}^{\infty} k\beta^{k-1} = \beta \sum_{k=0}^{\infty} \frac{d}{d\beta}\beta^k = \beta \frac{d}{d\beta} \sum_{k=0}^{\infty} \beta^k = \beta \frac{d}{d\beta} \frac{1}{1-\beta} = \beta \frac{1}{(1-\beta)^2} = \frac{\beta}{\alpha^2}$

$$\therefore E(S_T) = (b_1 + b_2 T)\alpha \cdot \frac{1}{\alpha} - b_2 \alpha \cdot \frac{\beta}{\alpha^2}$$

$$= b_1 + b_2 T - \frac{\beta}{\alpha} b_2$$

$$= E(x_T) - \frac{\beta}{\alpha} b_2 \qquad (2.4.1)$$

Thus the smoothed statistic is a biased forecast for a linear trend process.

2.5 Double Exponential Smoothing (for linear trend process)

Recall simple exponential smoothing (for constant) from Ch. 1.

$$S_T = S_{T-1} + \alpha[x_T - S_{T-1}] = \alpha x_T + (1-\alpha)S_{T-1}$$

which is unbiased for constant b but it is biased for linear trend process. In double exponential smoothing, we let

$$S_T^{[2]} = \alpha S_T + (1-\alpha)S_{T-1}^{[2]} \qquad (2.5.1)$$

and it can be shown that

$$E(S_T^{[2]}) = E(S_T) - \frac{\beta}{\alpha} b_2 \qquad (2.5.2)$$

thus

$$\frac{\beta}{\alpha} b_2 = E(S_T) - E(S_T^{[2]})$$

$$b_2 = \frac{\alpha}{\beta}\left[E(S_T) - E(S_T^{[2]})\right] \qquad (2.5.3)$$

A logical estimate of b_2 at the end of time T is

$$\hat{b}_2(T) = \frac{\alpha}{\beta}(S_T - S_T^{[2]}) \qquad (2.5.4)$$

From (2.4.1), $E(S_T) = E(x_T) - \frac{\beta}{\alpha} b_2$

$$E(x_T) = E(S_T) + \frac{\beta}{\alpha} b_2$$

$$= E(S_T) + \frac{\beta}{\alpha} \cdot \frac{\alpha}{\beta}\left[E(S_T) - E(S_T^{[2]})\right]$$

$$= 2E(S_T) - E(S_T^{[2]}) \qquad (2.5.5)$$

Hence a reasonable estimate for x_T at the end of time T is

$$\hat{x}_T = 2S_T - S_T^{[2]} \qquad (2.5.6)$$

For our exponential smoothing (for linear trend)

$$x_t = b_1 + b_2 t + \varepsilon_t$$

thus we have (p.8)

$$\hat{x}_T = \hat{b}_1(T) + \hat{b}_2(T) \cdot T$$

$$\Rightarrow \hat{b}_1(T) = \hat{x}_T - \hat{b}_2(T) \cdot T$$

$$\Rightarrow \hat{b}_1(T) = \underbrace{2S_T - S_T^{[2]}}_{(2.5.6)} - T \cdot \underbrace{\frac{\alpha}{\beta}\left(S_T - S_T^{[2]}\right)}_{(2.5.4)} \qquad (2.5.7)$$

Initial estimates satisfy

$$S_0 = \hat{b}_1(0) - \frac{\beta}{\alpha}\hat{b}_2(0)$$

[from (2.4.1) $S_T = b_1 + b_2 T - \frac{\beta}{\alpha} b_2$

as $T=0 \Rightarrow S_0 = \hat{b}_1(0) + \hat{b}_2(0) \times 0 - \frac{\beta}{\alpha}\hat{b}_2(0)$

$= \hat{b}_1(0) - \frac{\beta}{\alpha}\hat{b}_2(0)$]

$$S_0^{[2]} = \hat{b}_1(0) - 2\frac{\beta}{\alpha}\hat{b}_2(0)$$

[from (2.5.1) and (2.4.1)

$$S_T^{[2]} = \underbrace{S_T}_{\text{(2.5.1)}} - \underbrace{\frac{\beta}{\alpha} b_2}_{\text{(2.4.1)}} = \underbrace{b_1 + b_2 T - \frac{\beta}{\alpha} b_2}_{\text{(2.4.1)}} - \underbrace{\frac{\beta}{\alpha} b_2}_{\text{(2.4.1)}} \overset{T=0}{\Rightarrow} S_0^{[2]} = \hat{b}_1(0) - 2\frac{\beta}{\alpha}\hat{b}_2(0)$$

]

where $\hat{b}_1(0)$ and $\hat{b}_2(0)$ are usually obtained by regression analysis on historical data.

Forecasting

For $$x_T(\tau) = x_{T+\tau} = b_1 + b_2(T+\tau)$$

$$= b_1 + b_2 T + b_2 \tau = E(x_T) + \tau b_2$$

then $$\hat{x}_T(\tau) = \underbrace{S_T + (S_T - S_T^{[2]})}_{(2.5.6)} + \tau \underbrace{\hat{b}_2(T)}_{(2.5.4)}$$

$$\hat{b}_2(T) = \frac{\alpha}{\beta}(S_T - S_T^{[2]})$$

$$= S_T + (S_T - S_T^{[2]}) + \tau \cdot \frac{\alpha}{\beta} \cdot (S_T - S_T^{[2]})$$

$$= S_T + \left(1 + \tau \frac{\alpha}{\beta}\right)(S_T - S_T^{[2]}) \qquad (2.5.8)$$

$\therefore$, $E[\hat{x}_T(\tau)] = b_1 + b_2(T+\tau)$

$\hat{x}_T(\tau)$ is the τ-step ahead forecast made at time T.

